

# Skills Summary

## Modeling with Systems and Matrices

Use this reference while completing your worksheet or when studying.

### Skill 1 — Turning a story into a system

**Rule:** one unknown quantity = one letter; one sentence of the story that compares quantities = one equation. You need as many independent equations as letters.

**Two useful patterns:** a *count* equation ( $x + y = \text{total number}$ ) and a *value* equation (price, mass or percent per item  $\times$  number = total).

**Solve by elimination:** scale one equation so a letter has matching coefficients, then subtract.

**Example:** 2 large boxes and 3 small boxes weigh 31 kg; 1 large box and 1 small box weigh 13 kg. Find each mass.

**Step 1.** Two unknown masses and two comparisons, so I want a 2-by-2 system: let  $x$  be the large mass and  $y$  the small mass, then eliminate one letter.

**Step 2.**  $2x + 3y = 31$  and  $x + y = 13$ . Times 2:  $2x + 2y = 26$ . Subtracting:  $y = 31 - 26 = 5$ , so  $x = 13 - 5 = 8$ .  $\Rightarrow$   **$x = 8$  kg large,  $y = 5$  kg small**

**Try it:** 3 large boxes and 2 small boxes weigh 31 kg; 1 large box and 1 small box weigh 12 kg. Find each mass.

check:  $3(8) + 2(5) = 24 + 10 = 34 \neq 31$

### Skill 2 — Solving a 2-by-2 system with a matrix

**Matrix form:**  $ax + by = e$  and  $cx + dy = f$  become  $AX = B$  with  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ ,  $X = \begin{bmatrix} x \\ y \end{bmatrix}$ ,  $B = \begin{bmatrix} e \\ f \end{bmatrix}$ .

**Determinant:**  $\det A = ad - bc$  (down-diagonal minus up-diagonal).

**Cramer's rule:**  $x = \frac{\det A_x}{\det A}$ ,  $y = \frac{\det A_y}{\det A}$ , where  $A_x$  and  $A_y$  are  $A$  with that variable's column replaced by  $B$ .

**Example:** Solve  $2x + y = 11$  and  $3x + 4y = 24$  with a matrix.

**Step 1.** The coefficients are messy for substitution but the system is square, so I put it in matrix form and use Cramer's rule.

**Step 2.**  $\det A = (2)(4) - (1)(3) = 5$ ;  $\det A_x = (11)(4) - (1)(24) = 20$ ;  $\det A_y = (2)(24) - (11)(3) = 15$ .  $\Rightarrow$   **$x = 4$ ,  $y = 3$**

**Try it:** Solve  $3x + 2y = 12$  and  $x + 4y = 14$  with a matrix.

check:  $3(2) + 2(3) = 6 + 6 = 12$ ;  $2 + 4(3) = 2 + 12 = 14$

(!) **Watch out:** if  $\det A = 0$  you cannot divide by it and  $A$  has no inverse. That is not a mistake — it means the two equations describe parallel lines (no solution) or the very same line (infinitely many). Compare the equations directly to say which.

### Skill 3 — Where a line meets a curve

**Rule:** solve the linear equation for one letter, substitute it into the nonlinear equation, and you are left with ONE quadratic equation in one letter.

**Then:** solve the quadratic, and put each root back into the *line* to get its partner coordinate. Two roots give two intersection points, one root gives a tangent touch, no real root means the graphs never meet.

**Example:** Where does  $y = x^2 - 2x - 3$  meet  $y = x + 1$ ?

**Step 1.** Both equations already give  $y$ , so I want the  $x$  values where the two outputs agree — set the right-hand sides equal and solve the quadratic that appears.

**Step 2.**  $x^2 - 2x - 3 = x + 1 \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x - 4)(x + 1) = 0$ , so  $x = 4$  or  $x = -1$ ; the line gives  $y = 5$  and  $y = 0$ .  $\Rightarrow$  **(4, 5)** and **(-1, 0)**

**Try it:** Where does  $y = x^2 + 2x - 1$  meet  $y = 3x + 1$ ?

(2, -1) and (1, 2) check